

SITE / FACILITY INFORMATION			
FACILITY:	Cadillac WWTP	Digestion Method: 1 (Anaerobic = 1, Aerobic = 2, Lime stabilized = 3) App. Method: Injection (Inj, Inc, Sfc, not case sensitive)	
MONTH & YEAR:	April 18, 2017		
FACILITY NPDES #	MI0020257		
MDEQ SITE ID #:	21N09W17-RW01		
COUNTY:	Wexford		
TOWNSHIP:	Clam Lake	Org. N-1 yr. ago: Org. N-2 yrs. ago: TOTAL ACRES: 40 ACRES USED: 25	
SECTION:	17		
OWNER:	Richard Wilton		
FARMER:	John Rydquist		
ADDRESS:	9897 S. 43 Rd.		
ADDRESS:	Cadillac, MI 49601		
TELEPHONE:	231-884-0719		
SUPERINTENDANT	Jeff Dietlin		
		App. Rate (Gal/Ac)	17,902
LATITUDE:	N44* 20'86"	App. Rate (DT/Ac)	1.91
LONGITUDE:	W85* 43'58"	Org. N App. (#/ac)	202

SOIL TEST RESULTS			
SAMPLE DATE:	07/27/16	NEXT CROP:	Corn
CEC:	6	CROP YLD GOAL:	160
PH:	6	NIT. REC. (lbs/ac):	190
LIME INDEX:	6.9	ADJ. N REC.	190
BRAY P1 (ppm):	85	AVAIL. N APP.	186
POTASSIUM (ppm):	69	SOIL & APP. P	324
CALCIUM (ppm):	825	P ₂ O ₅ REC. (lbs/ac):	0
MAGNESIUM (ppm):	61	K ₂ O REC. (lbs/ac):	70
		LIME REC. (T/ac):	1 ta

[illegible]

ZINC (Zn)	8.29	lbs/acre
------------------	------	----------

	BIOSOLIDS ANALYSIS	
Lab ID	P&N	703913
Sample Name		Cadillac WWTP
Date Collected		3/28/2017
Matrix		Sludge
Density	lbs/gal	8.45
Mercury	mg/kg	0.35
Nitrogen,Ammonia	mg/kg	38,100.00
Nitrogen,Total	mg/kg	0.00
Nitrogen,Total Avail.	lb/ton	0.00
Nit.,Total Kjeldahl	mg/kg	91,200.00
pH	STD Units	8.10
Total Phosphorus	mg/kg	40,400.00
Nitrogen,Nitrate	mg/kg	20.00
Cadmium	mg/kg	2.50
Calcium	mg/kg	29,300.00
Chromium	mg/kg	97.20
Copper	mg/kg	1,020.00
Lead	mg/kg	31.40
Magnesium	mg/kg	5,220.00
Molybdenum	mg/kg	25.00
Nickel	mg/kg	29.30
Potassium	mg/kg	4,170.00
Zinc	mg/kg	1,480.00
Arsenic	mg/kg	26.60
Selenium	mg/kg	6.30
Total Solids	%	2.52
Total Volatile Solids	%	57.20

BIOSOLIDS APPLIED

[illegible]

Source:	Cadillac WWTP	Owner:	Richard Wilton	App. Rate (Gal/Ac)	17,902
Date:	04/18/17	Farmer:	John Rydquist	App. Rate (DT/Ac)	1.91
NPDES #:	MI0020257	Address:	9897 S. 43 Rd.	Useable Acres	40.0
MDEQ Site #:	21N09W17-RW01	City:	Cadillac, MI 49601	Acres Applied	25.0
County:	Wexford	Telephone:	231-884-0719	Seasons Used	4
Township:	Clam Lake	Latitude:	N44° 20'86"		
Section:	17	Longitude:	W85° 43'58"		

BIOSOLIDS ADDITIONS									
<u>Nutrient</u>	<u>Biosolids Additions</u>		<u>Residual Organic N Credit</u>		<u>Soil Fertility Test</u>		<u>Total Estimated Nutrients</u>		
Nitrogen (lb N/Ac)	186	+	0	+	0	lb N	=	186	lb N/Ac
Phos (lb P/Ac)	154			+	170	lb P	=	324	lb P/Ac
Pot (lb K/Ac)	16			+	138	lb K	=	154	lb K/Ac

SOIL ANALYSIS AND CROP INFORMATION					
C.E.C.:	6.0	P (lbs/acre):	170	K (lbs/acre):	138
Soil pH:	6.0	P (ppm):	85	K (ppm):	69
Lime Index:	6.9	Ca (lbs/acre):	1650	Mg (lbs/acre):	122

FERTILIZER RECOMMENDATIONS (lbs./a)						
Crop to be Fertilized	Yield Goal	N	P2O5	K2O	Lime	
-----	-----	-----	-----	-----	-----	
Corn	160	190	0	70	1 ta	Lbs./A

BIOSOLIDS ANALYSES									
Date: 28-Mar		Dry wt. basis	Lbs/dry ton	Lbs Applied Per Acre	Allowable Lifetime Lbs/acre	Previous Applied lb/acre	Total Applied lb/acre	Life in Years	% Cum.
		-----	-----	-----	-----	-----	-----	-----	-----
Density	(mg/l)-->	1.01	----	----					
Weight	(Lb/Gal)>	8.45	----	----					
Solids	(%)----->	2.52	----	----					
TKN	(%)----->	9.12	----	----					
Amm. N	(%)----->	3.81	----	----					
Nit. N	(%)----->	0.002	----	----					
Total Plant Avail. N----->			97	186					
Total P	(%)--->	4.04	81	154					
Total K	(%)--->	0.42	8	16					
Total Ca	(%)--->	2.93	58.60	111.69					
Total Mg	(%)--->	0.52	10.44	19.90					
Total As	(ppm)->	26.60	0.053	0.101	37	0.14	0.24	365	0.65%
Total Cd	(ppm)->	2.50	0.005	0.010	35	0.05	0.06	3,673	0.17%
Total Cr	(ppm)->	97.20	0.194	0.371	N/A	0.56	0.93	N/A	N/A
Total Cu	(ppm)->	1020.00	2.040	3.888	1,335	5.39	9.28	343	0.70%
Total Pb	(ppm)->	31.40	0.063	0.120	267	0.31	0.43	2,231	0.16%
Total Hg	(ppm)->	0.35	0.001	0.001	15	0.00	0.00	11,242	0.01%
Total Mo	(ppm)->	25.00	0.050	0.095	N/A	0.16	0.26	N/A	N/A
Total Ni	(ppm)->	29.30	0.059	0.112	274	0.17	0.28	2,453	0.10%
Total Se	(ppm)->	6.30	0.013	0.024	89	0.05	0.07	3,706	0.08%
Total Zn	(ppm)->	1480.00	2.960	5.642	2,492	8.29	13.93	442	0.56%

Month/Year: April-17																Michigan Department of Environmental Quality												MDEQ Site # 21N09W17-RW01				CROP & SOIL DATA			
Plant ID: MI0020257																Water Division												# of seasons utilized:				4		Crop to be fertilized:	
Superintendent: Jeff Dietlin																Biosolids Recycling Tracking Sheet												Acres utilized this month:				25		Corn	
Facility Name: Cadillac WWTP				Use of this form is voluntary. Submission of this form is not required																								Total acres in site:				40		Subsequent Crop:	
DAY	BIOSOLIDS APPLIED				NUTRIENTS								METALS																						
	MG	Solids %	Vol. %	Dry Tons	TKN %	NH4 %	NO3 %	TKN lb/ac	Phosphorus % lb/ac	Potassium % lb/ac	Arsenic mg/kg lb/ac	Cadmium mg/kg lb/ac	Copper mg/kg lb/ac	Lead mg/kg lb/ac	Mercury mg/kg lb/ac	Molybdenum mg/kg lb/ac	Nickel mg/kg lb/ac	Selenium mg/kg lb/ac	Zinc mg/kg lb/ac																
1	0.085	2.5		9.076355	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Bray P1 (ppm):				
2	0.149	2.5		15.88362	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		85				
3	0.057	2.5		6.050903	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		K (ppm):				
4	0.156	2.5		16.63998	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		69				
5	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Crop Yield Goal:				
6	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		160				
7	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Nitrogen Recom. (lb/ac)				
8	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		190				
9	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480						
10	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		CPLR METAL				
11	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		ACCUMULATIONS				
12	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Metal Total Maximum				
13	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Lead 0.43 37				
14	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Zinc 13.93 2495				
15	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Copper 9.28 1335				
16	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Nickel 0.28 374				
17	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Cadmium 0.06 35				
18	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Arsenic 0.24 37				
19	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Mercury 0.00 15				
20	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Molybdenum 0.26 NA				
21	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		Selenium 0.07 89				
22	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480						
23	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480		REMARKS				
24	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480						
25	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480						
26	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480						
27	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480						
28	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480						
29	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480						
30	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480						
31	0.000	0.0		0	9.12	3.81	0.002		4.04		0.42		26.6		2.5		1020		31.4		0.4		25.0		29.3		6.3		1480						
Average		2.52			9.12	3.81	0.002		4.04		0.42		26.60		2.5		1020		31.4		0.35		25.0		29.3		6.30		1480						
Totals	This Mo	0.448		47.7				186		154.0		15.9		0.1		0.0		3.9		0.1		0.0		0.1		0.0		5.6		BIOSOLIDS ANALYSIS DATE: 3/28/2017					
	This Yr																												SOIL ANALYSIS DATE: 7/27/2016						
	Cumulative													0.24		0.06		9.28		0.43		0.00		0.26		0.28		0.07	13.93						

42843

Total acres in site: 40

K 138 lbs/a

[illegible]

[illegible]

Org. N-1 yr. Ago (lbs/ac):	0	0	Residual N Credit
Org. N-2 yrs. Ago (lbs/ac):	0	0	Residual N Credit

3850.19	397.408	2.53503	0.238254	9.263327	97.20776	2.992474	0.033356	2.382543
---------	---------	---------	----------	----------	----------	----------	----------	----------

As	37
Cd	4.5
Cu	150
Pb	268
Hg	15
Ni	60
Se	89
Zn	300

[illegible]

29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
0.112		0.024		5.642		111.694		19.899		Err:509
0.112		0.024		5.642		111.694		19.899		Err:509
0.125		0.027		6.319		125.097		22.287		Err:509
0.282		0.074		13.932		111.694		19.899		Err:509
0.315		0.083		15.604		125.097		22.287		Err:509

0.17	0.05	8.29	0	0	0
------	------	------	---	---	---

[illegible]

[illegible]

Application Date	Facility	Facility #	Site I D	County	Township	Section
April 18, 2017	Cadillac WWTP	MI0020257	21N09W17-RW01	Wexford	Clam Lake	17

Owner	Farmer	Farmer Street Ad	Farm City, St.	Zip	Frm Telephone	Entity Super	Latitutde
Richard Wilton	John Rydquist	9897 S. 43 Rd.	Cadillac, MI	49601	231-884-0719	Jeff Dietlin	N44* 20'86"

Longitude	Season Used	Dig Method	App Method	Ttl Acres	Acres Used	Soil Date
W85* 43'58"	4	1	Injection	40	25	7/27/2016

CEC	Ph	Lime Index	Bray P1	Pot	Calcium	Magnesium
6	6	6.9	85	69	825	61

Intend Crop	Yld Goal	N Rec	Phos Rec	Pot Rec	Lime Rec
Corn	160	190	0	70	1 ta

Facility	Facility #	Lab I.D.	Lab Ref #	Sample ID	Date Collected	Matrix	Density
Cadillac WWT	MI0020257	P&N	703913	Cadillac WW	3/28/2017	Sludge	8.45

Hg	N Amonia	N Total	N Ttl.AvaIL #/T	N Ttl Kjeldahl	pH	Ttl. Phos	N Nitrate
0.35	38100	0	0	91200	8.1	40400	20

Cd	Ca	Cr	Cu	Pb	Mag	Mo	Ni
2.5	29300	97.2	1020	31.4	5220	25	29.3

K	Zn	As	Se	Ttl. Solids	Ttl. Vol. Solids
4170	1480	26.6	6.3	2.52	57.2

Facility	Facility #	Site ID	Date 1	Gallons 1	% Solids 1	Date 2	Gallons 2
adillac WW	MI0020257	N09W17-RW	4/18/2017	85,248	2.52	4/19/2017	149,184

% Solids 2	Date 3	Gallons 3	% Solids 3	Date 4	Gallons 4	% Solids 4	Date 5
2.52	4/24/2017	56,832	2.52	4/26/2017	156,288	2.52	

Gallons 5	% Solids 5	Date 6	Gallons 6	% Solids 6	Date 7	Gallons 7	% Solids 7
-----------	------------	--------	-----------	------------	--------	-----------	------------

Date 8	Gallons 8	% Solids 8	Date 9	Gallons 9	% Solids 9	Date 10	Gallons 10
--------	-----------	------------	--------	-----------	------------	---------	------------

% Solids 10	Date 11	Gallons 11	% Solids 11	Date 12	Gallons 12	% Solids 12	Date 13
-------------	---------	------------	-------------	---------	------------	-------------	---------

Gallons 13	% Solids 13	Date 14	Gallons 14	% Solids 14	Date 15	Gallons 15	% Solids 15
------------	-------------	---------	------------	-------------	---------	------------	-------------

Date 16	Gallons 16	% Solids 16	Date 17	Gallons 17	% Solids 17	Date 18	Gallons 18
---------	------------	-------------	---------	------------	-------------	---------	------------

% Solids 18	Date 19	Gallons 19	% Solids 19	Date 20	Gallons 20	% Solids 20	Date 21
-------------	---------	------------	-------------	---------	------------	-------------	---------

Gallons 21	% Solids 21	Date 22	Gallons 22	% Solids 22	Date 23	Gallons 23	% Solids 23
------------	-------------	---------	------------	-------------	---------	------------	-------------

Date 24	Gallons 24	% Solids 24	Date 25	Gallons 25	% Solids 25	Date 26	Gallons 26
---------	------------	-------------	---------	------------	-------------	---------	------------

% Solids 26	Date 27	Gallons 27	% Solids 27	Date 28	Gallons 28	% Solids 28	Date 29
-------------	---------	------------	-------------	---------	------------	-------------	---------

Gallons 29	% Solids 29	Date 30	Gallons 30	% Solids 30	Date 31	Gallons 31	% Solids 31
------------	-------------	---------	------------	-------------	---------	------------	-------------